DawaLink

A local-first Moroccan medicine enquiry assistant with conversational AI

Technical report · 2026-10-27 · repository baseline f00355a

Implemented scope

DawaLink guides a customer through a real pharmacy question: describe a stock or price need in Darija, Arabic or French, confirm the matching medicine and exact strength in the local catalogue, then copy a message for a pharmacy. Qwen3 1.7B runs locally through Ollama and extracts intent and details into a constrained schema. SQLite supplies all product facts. Optional image OCR uses Tesseract locally. No hosted AI service or paid API is used.

Observed live integration flow: the Qwen model parsed a Darija stock question with a requested city, returned three 1 G catalogue candidates with correct package sizes, and the selected catalogue ID generated an editable Darija enquiry. A separate Arabic symptom question received the safety response. These are smoke interactions, not a benchmark. A previous core suite had 18 tests pass; the new LLM flow has no representative accuracy evaluation.

This is a pharmacy enquiry aid, not clinical decision support. It does not diagnose, prescribe, verify stock, find nearby pharmacies, contact an office, or send user text to a cloud service. Prompts go to local Ollama on loopback.

1. Product and architecture

The user speaks in their own words, and the local Qwen model extracts goal, medicine, strength, form, quantity and city as JSON. A grounding step rejects details not supported by the message. Candidate records are retrieved and ranked from SQLite; strength and stated form conflicts are filtered. For an image, the API validates JPEG/PNG/WebP, applies local preprocessing, runs Tesseract, then searches the same catalogue. A pharmacy enquiry requires the user to select a real catalogue ID.

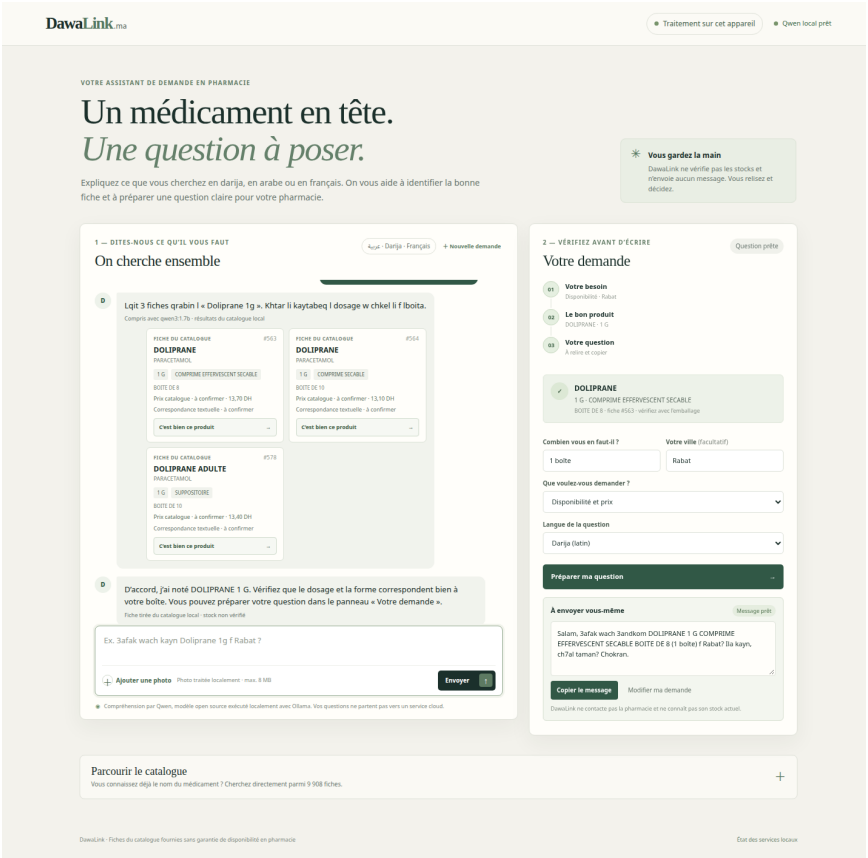
Runtime flow

Browser chat	Python WSGI API	Local model + core	SQLite + FTS5
Describe · confirm · copy	validation · grounded results	Ollama · Qwen3 1.7B · OCR	9,908 catalogue records

Customer story

A customer asks in Darija whether a medicine is available in Rabat. The assistant understands the goal, finds three exact 1 G records, and asks the customer to confirm the package. After the choice, the customer selects the question type, edits quantity or city, reviews the message and copies it to contact a pharmacy. Pharmacy inventory, price changes and replies remain with the real pharmacy.

Figure 1. Live browser workflow: a Darija request, Qwen interpretation, exact catalogue options, confirmed product, and the message ready for the customer to copy.



2. Data source and quality

The supplied `moroccan_medications.csv` is the only imported dataset. It contains 9,908 rows and ten source columns: id, specialite, substance_active, dosage, forme, presentation, epi, statut, ppv and ph. All IDs are unique; 4,070 distinct brand strings were observed. Source export date was not recorded, and the supplied `ammps_cards.csv` is empty.

Missing/unavailable values are preserved, not imputed. EPI is blank for all rows; status is blank for 1,810; PPV and PH each have 1,098 empty or explicit no-price values. No manufacturer information is recoverable from the all-empty EPI field.

Missing or unavailable field	Rows	Share of 9,908
EPI	9,908	100.0%
Status	1,810	18.3%
PPV	1,098	11.1%
PH	1,098	11.1%

Source investigation

The official AMMPS search page was reachable and reported 9,936 entries with an update date of 2026-10-01 during research. The older repository export has 9,908 rows; it is not presented as current or complete. AMMPS bulk reuse terms were not identified, so the application does not scrape or refresh that site.

The Moroccan Open Data portal lists a CNOPS medicines reference under ODbL, but the linked XLSX is from 2014. The portal returned 503 and its old resource host did not resolve in this environment. The dataset is not mixed into current catalogue records because it could not be fetched and its prices/status are stale. The original repository has no explicit data license file; check reuse rights before redistribution.

Normalization stores original text plus deterministic normalized brand/substance keys, source attribution and import timestamp. SQLite rebuild writes a new temporary database and swaps it in after import. FTS5 and indexes support local ranked lookups. `scripts/ingest_data.py` emits a JSON quality report. The raw CSV remains unchanged.

3. Search and request understanding

Search combines SQLite FTS5, normalized text, partial matching and Python `SequenceMatcher`. The model extracts product name, strength and form from the conversation, but the application grounds those fields against what the user said and the available catalogue. Explicit strength and form conflicts are filtered. Scores are matching signals, not calibrated probabilities. The user confirms the exact product before an enquiry can be generated.

The conversational endpoint sends the last user turns to Qwen running at `127.0.0.1:11434` and requests schema-constrained JSON. It handles French, Arabic, Darija Latin/Arabizi, code switching, follow-up turns, and asks for a product name when it is missing. User-stated package quantity stays separate from package size in a catalogue record. A high-precision local guard handles obvious symptom and personal-dosing requests; no model answer is allowed to prescribe or invent availability.

Query	Observed top result	Top 3 target hit
Doliprane 1 G	DOLIPRANE · 1 G	yes
doliprene	DOLIPRANE · 1 G	yes
AMOXICILLINE	ALFAMOX · 1 G	yes
Amlodipine	AM-10 · 10 MG	yes
DOLIPRANE 500 MG	DOLIPRANE · 500 MG	yes

Observed live result · smoke interaction

```
Darija: “3afak wach kayn Doliprane 1g f Rabat?”
intent=availability · product=DOLIPRANE · strength=1 G · city=Rabat
3 exact-strength candidates returned; the user selected ID 563 (BOITE DE 8).
```

If no name can be identified, DawaLink asks for clarification rather than filling a medicine name from generic words like ‘pharmacy’ or ‘children’.

4. OCR and image matching

The OCR executable is Tesseract 5.5.0 in the local verification environment; English, French and Arabic trained data were installed. OCR is optional. There is no OCR cloud upload and no vision-language model. Files are capped at 8 MB, checked by file signature and pixel count, preprocessed locally, written to a temporary file and deleted after inference. Text matching uses the same SQLite catalogue.

The automated suite generated a text image and verified that Tesseract read `DOLIPRANE 1 G` and matched actual catalogue records. This proves pipeline wiring, not photograph performance.

Real-image spot check

One test image was retrieved from the MIT-licensed Medicine Boxes Annotated dataset (test split). The dataset annotation labels only a generic medicine box; the prominent visible product name DiaCURE was manually transcribed. The product is not the Moroccan D-CURE record. Local OCR extracted ENFANTS / CHILDREN but missed DiaCURE: visible-text CER was 31.8%; product-name read 0/1; Moroccan catalogue top-1/top-3 identification 0/1. The app returned no candidate and required confirmation, avoiding a false D-CURE suggestion. This single foreign-market image is a failure case, not a representative benchmark.

A positive real-photo identification check still needs a photograph of a real product present in the included Moroccan catalogue, with its expected catalogue ID. The image must show a readable brand and strength. This is the only external input needed to close the photo acceptance check.

Image source: Medicine Boxes Annotated, MIT license as listed on dataset card; 874 images, 87.3 MB, one generic object class. The raw image is downloaded only by the optional evaluation script and is not redistributed here.

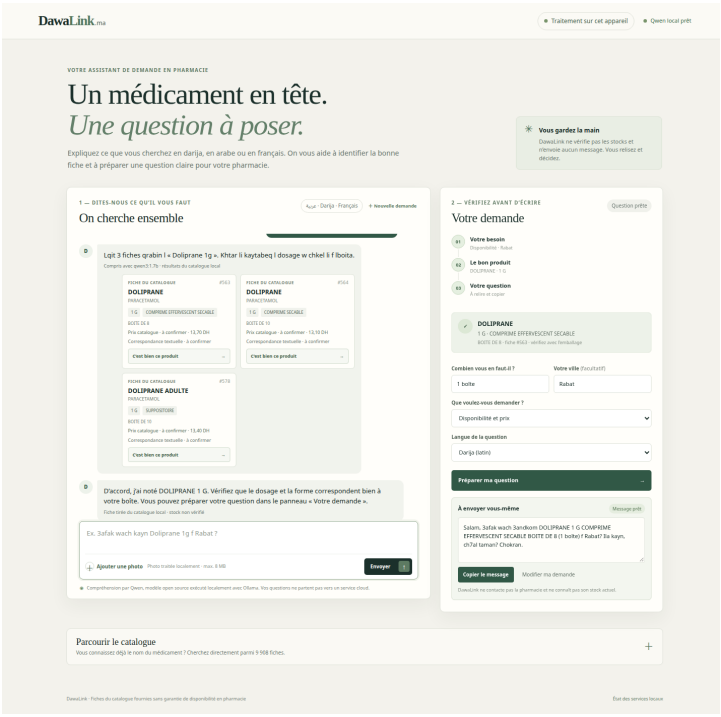
5. API, storage and pharmacy workflow

Endpoint	Purpose
GET /api/health	record count, OCR status, unknown export date
GET /api/medications?q=…	ranked catalogue search and paging
GET /api/medications/{id}	one real catalogue record
POST /api/assistant/message	local LLM dialogue, grounded candidates and next step
GET /api/assistant/status	Ollama and model availability
POST /api/interpret	deterministic compatibility parser
POST /api/recognize	validated multipart image OCR and candidates
POST /api/pharmacy-request	message from a verified catalogue ID
GET /api/export	CSV export of search results

Database rows preserve the original ten source fields with brand_key, substance_key, source and imported_at; id is the primary key. FTS5 indexes brand, ingredient, dosage, form and presentation. Filter/search SQL is parameterized. Uploaded bytes are not persisted.

The enquiry API refuses a name without a valid local catalogue ID. The UI allows the user to edit dosage, form and package presentation, to select message language, quantity and location, then copy the generated text. Quantity is not confused with package count. No availability is asserted and no message is sent.

Figure 2. Live generated enquiry after product selection. The user sends the message independently.



6. Evaluation and test results

The previously run core automated suite passed 18 tests with local Tesseract configured. Separately, the current LLM route was exercised over HTTP with a Darija availability request, an Arabic symptom question, and a French price question. The full UI workflow was captured from the running app through candidate selection and message copy. The new model route has only smoke coverage; no representative language or entity-accuracy score is claimed.

Intent accuracy: 100.0% on 16 manually authored cases. Language labels: 100.0%. Named product extraction: 7/7. Clarification: 16/16. Each intent had only 1–6 examples; the set is not representative or independently reviewed, so 100% is only a smoke/evaluation score on these examples.

Evaluation component	Set size	Observed result
Intent/language rules	16 authored examples	16/16 each; narrow test set
Catalogue searches	6 authored probes	6/6 top-three hits
Strength conflict	1 query, top 3	all 3 match 500 mg
Ambiguous brand	DOLIPRANE top 20	7 strengths; selection required
Generated-image OCR smoke	1 synthetic image	text read; not photo evidence
Real-photo OCR	1 foreign product photo	name missed; safe no-match

Detailed machine-readable results: `evaluation_results.json` and `image_evaluation_results.json`. Full results are reproducible with `scripts/evaluate.py` and `scripts/evaluate_public_ocr.py`.

7. Reproduction, safety and limitations

Runtime uses Python standard library and SQLite, plus the separately installed Ollama runtime and one-time Qwen3 1.7B weights for conversation. OCR uses local Tesseract and language packs; optional image preprocessing uses Pillow. Deliverable generation uses the separate pinned requirements file.

Launch from the repository root

```
ollama pull qwen3:1.7b
python3 scripts/ingest_data.py
python3 app.py
```

Start `ollama serve` first if Ollama is not running as a service, then open http://127.0.0.1:8000. Model download is about 1.4 GB and is not bundled.

Verification commands

```
python3 -m unittest discover -s tests -v
python3 scripts/evaluate.py
python3 scripts/evaluate_public_ocr.py
```

Safety limitations

No diagnosis, personal dosing instructions, treatment suggestions, substitutions, interaction checker, pregnancy or child-safety decisions, pharmacy availability integration, pharmacy directory, voice transcription, or external messaging exists. OCR match scores are not probabilities. Confirmation is mandatory for ambiguous results. Catalogue price/status are historic source fields, not live pharmacy data. The local 1.7B model can misunderstand Darija; its parse is constrained and product choices are grounded in the catalogue, but user confirmation remains necessary.

References

[1] AMMPS official medicine register. [2] Morocco Open Data reuse license (ODbL-derived). [3] CNOPS medicine reference listing. [4] Tesseract OCR source. [5] Medicine Boxes Annotated dataset. [6] Darija Open Dataset (investigated, not used for pharmacy-intent training). [7] Ollama local chat API and structured outputs. [8] Qwen3 model and Apache 2.0 license.